

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I V. RUPASHREE (192319012) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 



Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.
3. A smart traffic management system is being designed to control traffic signals across a city based on vehicle density, pedestrian movement, and weather conditions. The system must respond to real-time sensor inputs while also learning traffic patterns over time to reduce congestion. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for minimizing congestion and travel time.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

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C01 - AT3

Name: V. RUPASHREE

Regno.: 192319012

Course code: CSA1710

Course name: Artificial
Intelligence

Smart Home Automation System using Intelligent Agent.

A smart home automation system controls lighting, temperature and security by continuously monitoring environmental conditions and user behaviour. Intelligent agents help the system perceive the environment, make decisions and perform actions automatically.

Simple Reflex Agent:-

A Simple Reflex agent acts only on the current percept without considering past information. It follows predefined condition-action rules.

Example: If motion is detected in a room $\rightarrow$ Turn ON lights.
If no motion is detected $\rightarrow$ Turn OFF lights.

Advantages:-

1. Fast Response
2. Easy to implement

Limitations:-

1. cannot learn user preferences
2. cannot handle partially observable environments.

Model Based agent:-

A agent that maintains an internal model of the environment and uses previous information to make decisions.

Example:-

- The system remembers that the living room was occupied recently and keeps light ON even if the motion is temporarily absent.
- Tracks temperature changes over time.

Advantages:-

1. Works efficiently in dynamic environments.
2. ~~Set~~ Maintains state information.

Limitations:-

1. More complex than reflex agent.
2. Requires memory and computation.

3. Goal-Based agent:-

A goal-based Agent selects actions may help achieve a specific goal.

Example:-

Goal: Maintain Room temperature at 24°C

If temperature rises above 24°C - Activate cooling system.

If temperature falls below 24°C - Activate heating system.

Advantages:-

1. Flexible decision making
2. Can evaluate alternative actions.

Utility-Based Agent:-

A utility based agent chooses actions that maximize overall satisfaction or utility.

Example:-

The system handles:-

- User comfort
- Energy consumption
- Security.

During night time:-

- Dim lights instead of full brightness.
- Maintain comfortable temperature while minimizing power consumption.
- Activate advanced security monitoring.

Advantages:-

- Handles multiple objectives simultaneously.
- Produces optimal decisions.

Limitations:-

- Utility function design can be complex.

Most Effective Approach:-

A Hybrid based Agent combining model based, goal based, and utility-based Agents is the most effective solution.

Justification:-

1. Comfort
2. Energy efficiency
3. Security.

2) Robot Navigation in an unknown Maze using UCS & IDS

A robot in an unknown maze must find the exit without prior knowledge. Uninformed search algorithms help explore the maze systematically.

UCS: Uniform Cost Search

- Expands the node with the lowest path cost first.
- Uses a priority queue
- Guarantees the optimal path.

Advantages:-

1. Complete and optimal
2. Suitable when path costs differ.

Disadvantages:-

1. Requires large memory.
2. Computationally expensive.

IDS - Iterative Deepening Search

1. Repeatedly performs depth limited search with increasing depth.
2. Combines DFS memory efficiency & BFS completeness.

Advantages:-

1. Low memory usage
2. Complete for finite search spaces.

Disadvantages:-

1. Repeat node expansions.
2. May take more time.

Relation to Intelligent Agents

- * Reflex agents are unuitable because they cannot remember employed paths.
- * Model based agents store more information
- * Goal based agents focus on reaching the exit.
- * Utility based agents choose the most efficient path.

Best combinations:-

A goal based Model-based Agent with UCS.

Effective solution because it remembers employed paths focuses on the goal and finds the lowest cost route to exit.

~~Conclusion:-~~